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Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

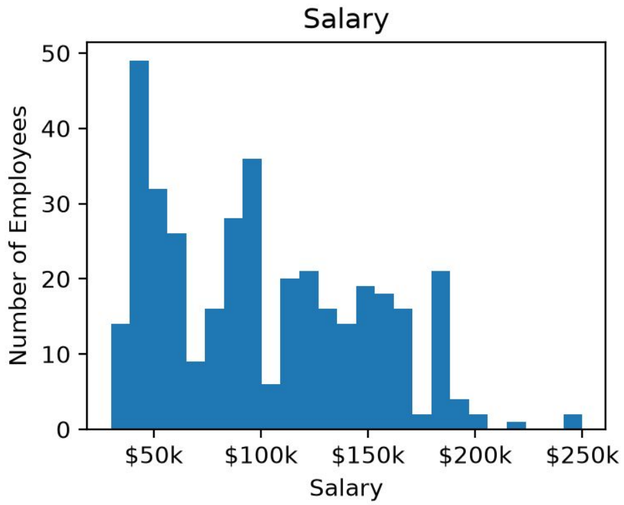
Data Description

373 records of employee demographic, experience, and salary data

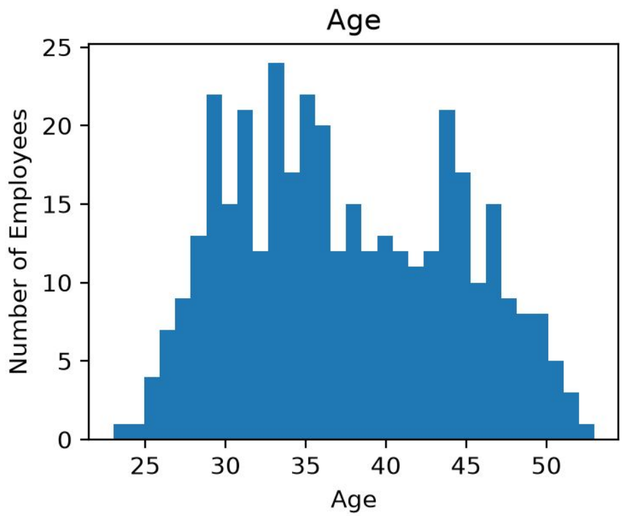
- Salary
- Age
- Years of Experience
- Gender
- Education Level
- Job Title

The dataset had no corrupt records. Empty rows and one outlier with a very low salary ($350) were the only deletions

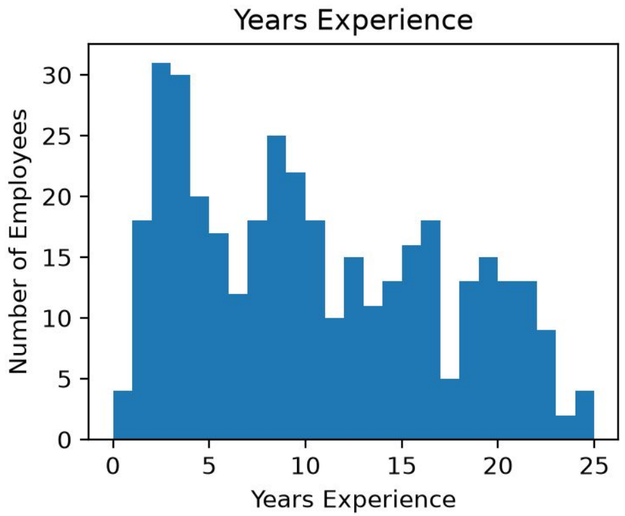
Data Description: Distributions



mean	$100,846.77
min	$30,000.00
25%	$55,000.00
50%	$95,000.00
75%	$140,000.00
max	$250,000.00

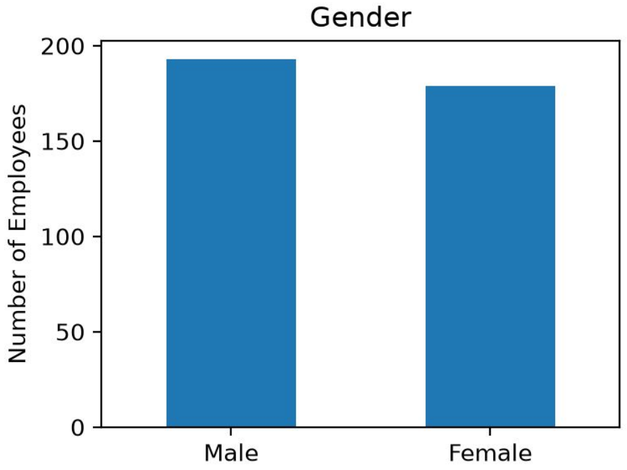


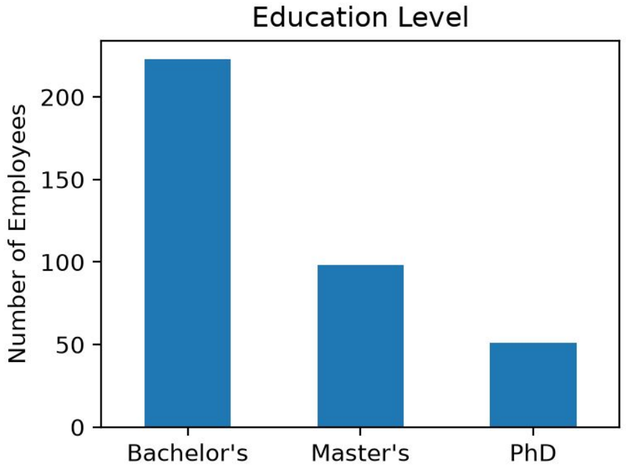
mean	37.454
min	23.000
25%	31.750
50%	36.000
75%	44.000
max	53.000



mean	10.054
min	0.000
25%	4.000
50%	9.000
75%	15.000
max	25.000

Data Description: Distributions





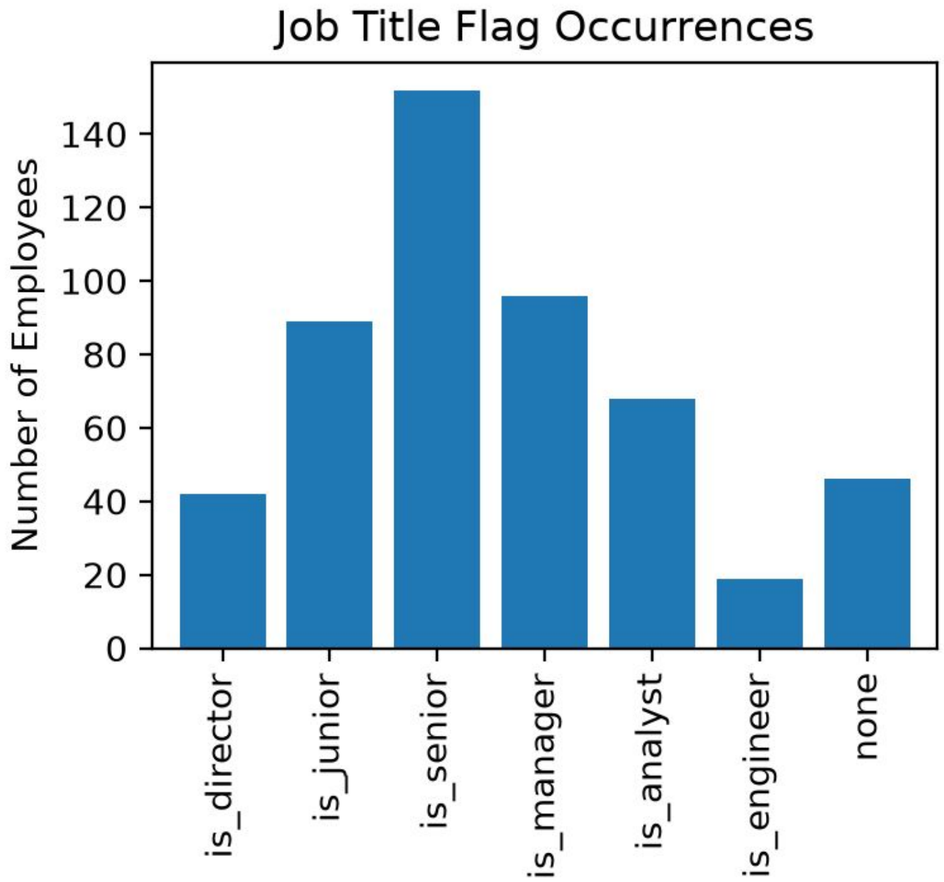
Male	193	51.9%
Female	179	48.1%

Bachelor’s	223	59.9%
Master’s	98	26.3%
PhD	51	13.7%

Data Description: Job Title Distribution

Due to the wide variety of job titles, job title data was broken down by keyword*

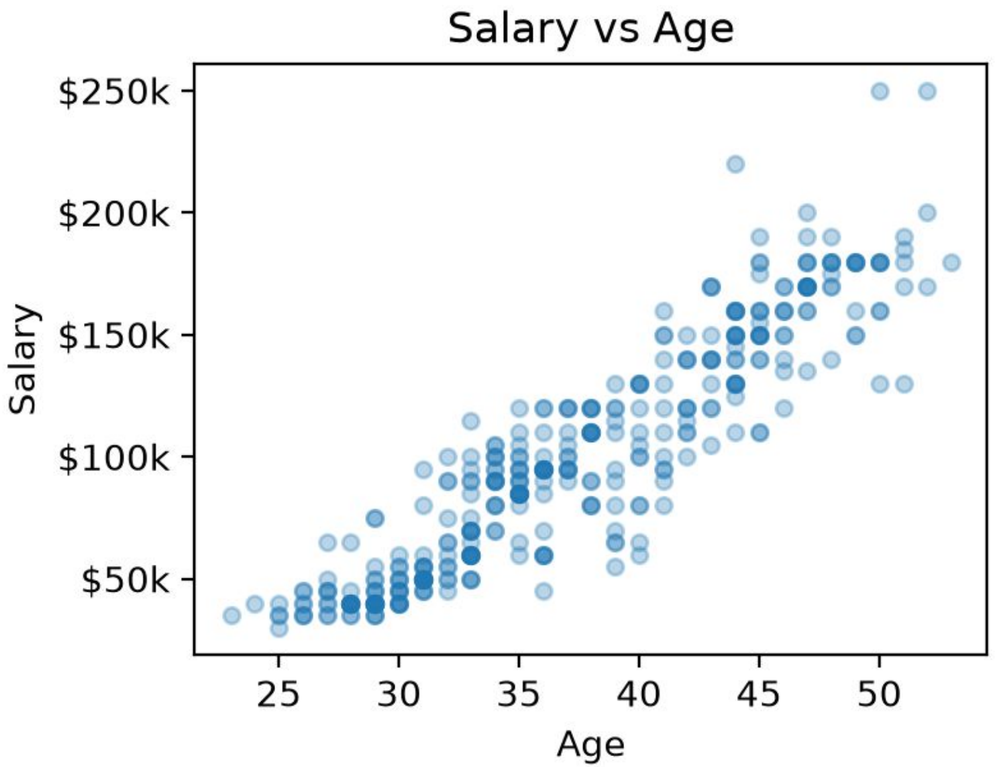
- “director”
- “junior”
- “senior”
- “manager”
- “analyst”
- “engineer”



*Some employees matched multiple keywords (e.g., “senior engineer”). 46 employees (12.37%) matched none

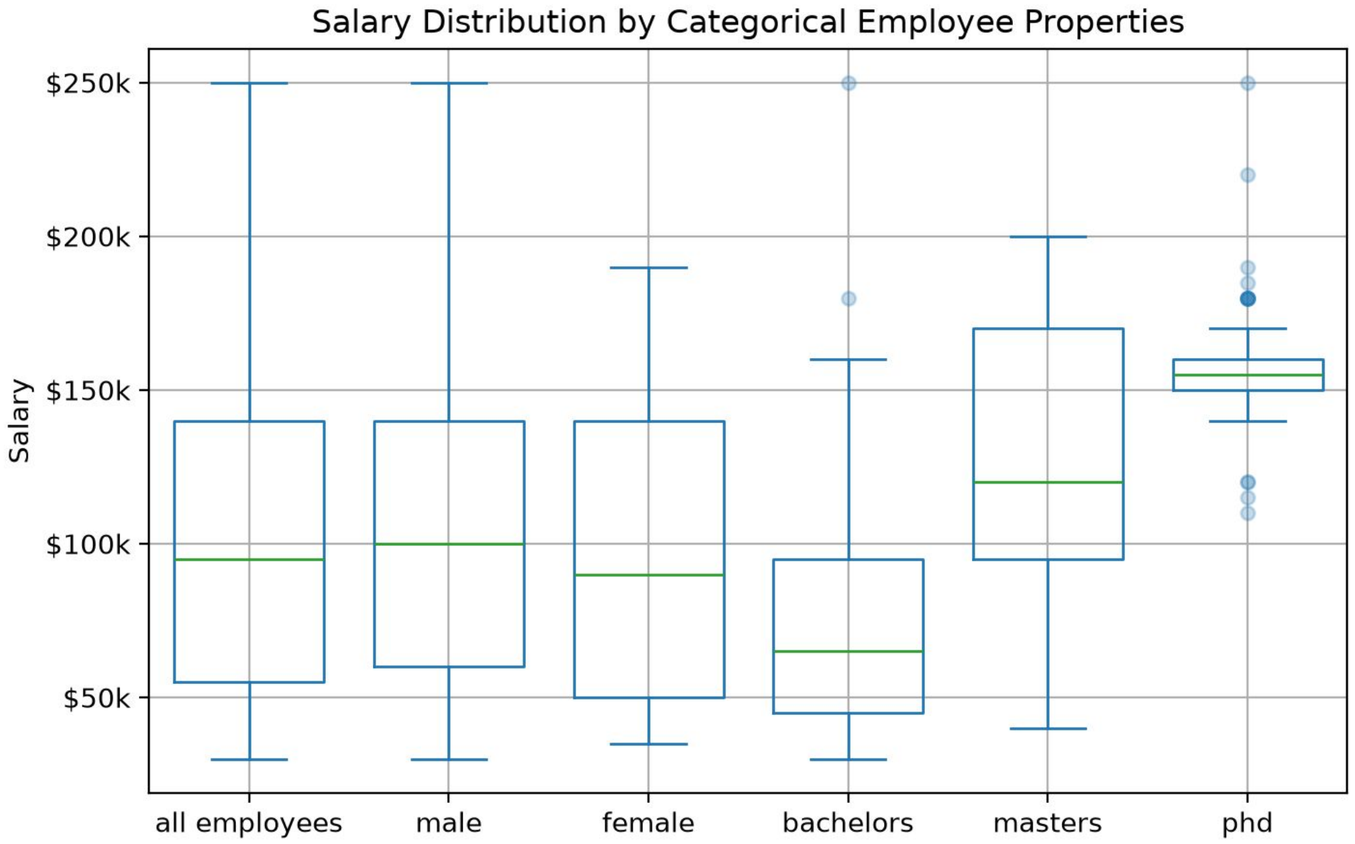
Data Description: Salary Correlation





Positive correlations are indicated between Years of Experience, Age and Salary. The correlation appears tighter and more linear for Years of Experience.

Data Description: Salary Correlation



	Mean		Median	
All Employees	$100,846.77		$95,000	

Gender	Mean	vs Male	Median	vs Male
Male	$104,404.15	–	$100,000.00	–
Female	$97,011.17	**-$7392.97**	$90,000.00	**-$10,000.00**

Education	Mean	vs Bachelors	Median	vs Bachelors
Bachelor's	$75,089.69	–	$65,000.00	–
Masters	$129,795.92	**+$54,706.23**	$120,000.00	**+$55,000.00**
PhD	$157,843.14	**+$82,753.45**	$155,000.00	**+$90,000.00**

Salary Linear Regression: Significant Variables

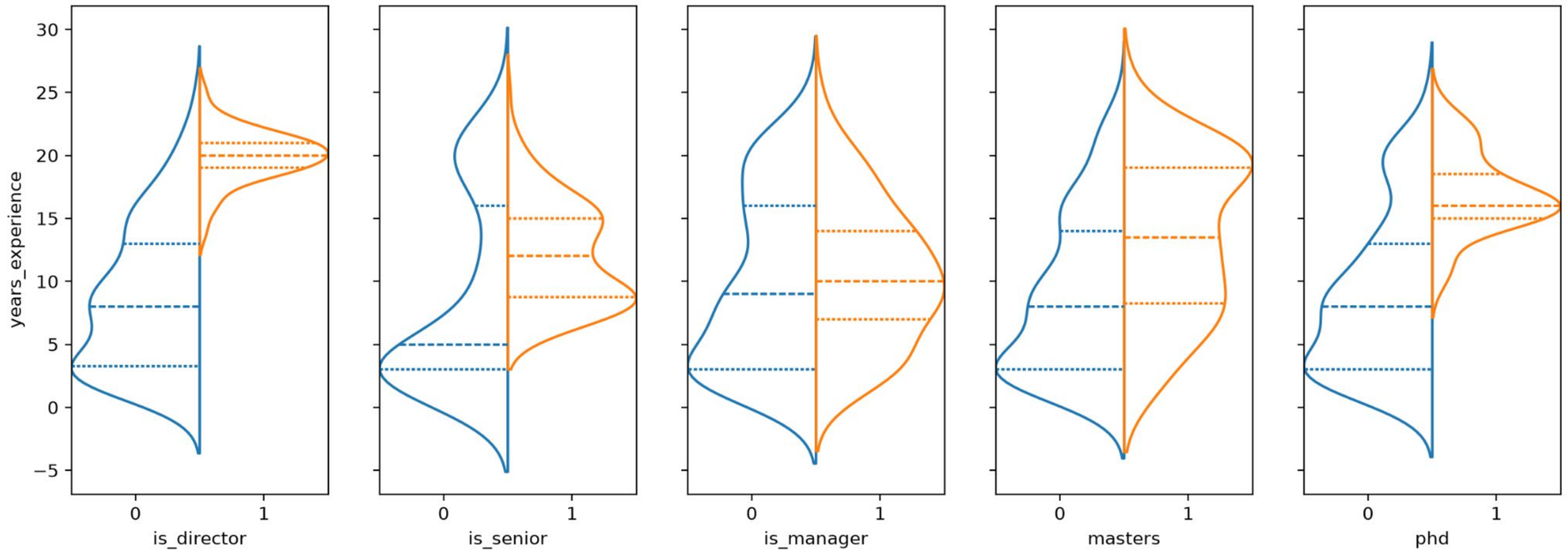
Preliminary linear regression to identify statistically significant predictor variables to salary.

Variable	P-Value	Is Significant
male	< 0.001	✅
years_experience	< 0.001	✅
masters	< 0.001	✅
phd	< 0.001	✅
is_director	< 0.001	✅
is_junior	**0.0885**	❌
is_senior	< 0.001	✅
is_manager	< 0.001	✅
is_analyst	**0.8460**	❌
is_engineer	**0.5013**	❌

Variables with p-value > 0.05 do not have a statistically significant association with Salary and will be omitted from the regression model.

Linear Regression: Years Experience Multicollinearity

Some variables show evidence of multicollinearity with Years of Experience.



Among these variables, True values in all but “is_manager” show distributions that are noticeably shifted towards higher Years of Experience values.

Linear Regression: Years Experience Multicollinearity

A linear regression of Years of Experience and variables suspected of multicollinearity indicated that all show a statistically significant relationship to variance of Years of Experience.

Suspected Variable	Coefficient	P-Value	VIF	omit from regression?
masters	2.6187	< 0.001	1.462968	no
phd	7.2576	< 0.001	1.650073	no
is_director	11.5747	< 0.001	2.251962	no
is_senior	3.7267	< 0.001	1.468272	no
is_manager	2.7672	< 0.001	1.166135	no

However, a variance inflation factor (VIF) test did not show any of the variables passing the > 10 or more stringent > 5 inflation factor threshold. This suggests our regression model would not be improved if we removed them.

Salary Linear Regression: Results

Variable	Coefficient	Std Err	P-Value	0.025	0.975
intercept	29068.37	1660.59	< 0.001	25802.81	32333.94
male	7848.84	1466.08	< 0.001	4965.80	10731.88
years_experience	5081.26	175.3	< 0.001	4736.53	5425.98
masters	13972.69	2008.27	< 0.001	10023.42	17921.95
phd	23137.04	2728.31	< 0.001	17771.82	28502.26
is_director	25052.49	3461.87	< 0.001	18244.71	31860.28
is_senior	14345.6	1799.29	< 0.001	10807.29	17883.92
is_manager	4174.95	1801.47	0.021	632.34	7717.55

R-Squared: 0.916

An Ordinary Least Squares linear regression model returned an R-squared value of 0.916 indicating a good fit, explaining 91.6% of the variance in Salary.

Interpretation and Next Steps

Years of Experience

For each year of experience, the expected salary of an employee increases $5,081.26.

The model reports 95% confidence the actual increase is between $4,736.53 - 5,425.98.

A 25 year career would be associated with $127,031.50 higher salary over a career that has just begun.



Interpretation and Next Steps

Male vs Female

Male employees have an associated increase in salary of $7848.84 over female employees.

The model reports 95% confidence the actual salary increase lies between $4965.80 - 10731.88.



Interpretation and Next Steps

Male vs Female

The model assumes the male/female salary difference remains constant regardless of years of experience, job title or other factors.

Further investigation could be done to look for non-linear behavior in the salary gap as experience and related factors increase.

